Could Spider-Man Really Turn Off His Powers?

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Small movie **spoilers** ahead! In Spider-Man: Brand New Day, Peter Parker has problems caused by spider **DNA** from the bite that gave him powers. He wants to turn off some spider traits and keep others. Bruce Banner, who becomes the Hulk, has built a superpower **inhibitor** that could help.

The movie idea is based on real science. Tiny **molecules** called siRNAs can turn off certain **genes**. **Genes** are instructions in our **DNA**. Cells copy those instructions and use them to build **proteins**. An siRNA grabs a copy and marks it to be chopped up, so the protein is not made. Doctors already use at least eight siRNA treatments for diseases.

Scientist Anastasia Khvorova says siRNAs are good at shutting down foreign **DNA**, like spider **DNA**. But Peter wants to stop several different traits, so he would probably need a mix of siRNAs. Scientist Judy Lieberman says no one can control siRNAs that precisely yet.

Word Treasure

spoilers -- Details that tell you what happens in a movie or story before you see it.
DNA -- The instructions inside every cell that tell a living thing how to grow and work.
inhibitor -- Something that blocks or slows something down.
molecules -- Very tiny pieces that make up everything.
genes -- Parts of DNA that hold instructions for things like eye color.
proteins -- Tiny parts made by cells that do many jobs in the body.

Background you need

In the movies, Peter Parker gets superpowers after a spider bites him, and he becomes Spider-Man.

Every living thing has DNA, which works like an instruction book for building and running its body.

Scientists have learned to make tiny molecules that can switch off certain instructions to help treat some diseases.

Break it down

WHO: Peter Parker in the movie, and scientists Anastasia Khvorova and Judy Lieberman
WHAT: A movie device that blocks powers is based on real molecules called siRNAs
WHERE: In Spider-Man: Brand New Day and in real science labs
WHY: siRNAs can turn off genes, so they might block spider DNA

Why it matters

A superhero movie can help you learn how real medicine works inside your own body.

Test what you remember

1. What is the name of the new movie?

(A) Spider Island
(B) The Hulk Returns
(C) Spider-Man: Brand New Day
(D) Web World

2. What does Peter Parker want to do?

(A) Move to a new city
(B) Turn off some spider traits
(C) Get more powers
(D) Become the Hulk

3. Who built a superpower inhibitor?

(A) Peter Parker
(B) A spider
(C) A doctor at a hospital
(D) Bruce Banner

4. What do siRNAs do?

(A) Make spiders
(B) Build houses
(C) Give people powers
(D) Turn off certain genes

5. How many siRNA treatments do doctors already use?

(A) At least eight
(B) None
(C) Exactly one
(D) Thousands

6. What does scientist Judy Lieberman say?

(A) siRNAs are dangerous candy
(B) No one can control siRNAs that precisely yet
(C) DNA is not real
(D) Spider-Man is real

Answer key (try the quiz first!): 1: C 2: B 3: D 4: D 5: A 6: B

Questions to think about

1. Anastasia Khvorova, scientist: Says siRNAs are good at shutting down foreign DNA, but Peter would need a mix of them.

2. Judy Lieberman, scientist: Says no one can control siRNAs that precisely yet.

MY THOUGHTS (at least 20 words):